MUHAMMAD SHAYAN AHMED

Machine Learning Engineer | AI Engineer
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PROFESSIONAL SUMMARY

Final-year Software Engineering student with experience in machine learning and AI engineering. Developed and evaluated machine learning models, including classification and predictive modeling, on large-scale datasets. Built full-stack AI applications using computer vision, generative AI, and large language models (LLMs). Completed internships in machine learning engineering and software development.

EDUCATION

Sir Syed University of Engineering & Technology (SSUET), Karachi *Expected July 2027*
Bachelor of Science in Software Engineering — Batch 2023F | CGPA: 3.48/4.00

EXPERIENCE

SWE Intern — KunjiSoft Technologies, Karachi (On-site) *Oct 2026 – Present*

- Developed responsive landing page and worked on production dashboard to resolve critical bugs and implement solutions, including mobile/tablet responsiveness, CSV export functionality, and real-time reporting features.
- Managed and handled production data across multiple factories, ensuring data handling, authenticity, and consistency; detected and fixed critical time inconsistency where dashboard and generated reports showed different timings by standardizing backend Firebase Firestore queries from UTC to Asia/Karachi (PKT).
- Analyzed and audited factory data through the dashboard, monitoring devices to verify correct data reporting and track production, idle, and off status.
- Briefly worked with edge CPUs detecting mold open/close events, power/electricity sensors, and environmental sensors (heat) to monitor production targets, and performed health checks of IoT devices to ensure data reliability and system uptime; built and deployed a Streamlit app to automate missing time report generation with CSV and PDF download, and integrated new Downtime Section with full CRUD using React.js.

ML Engineering Intern (Remote) — FlyRank AI *Jun 2026 – Sep 2026*

- Built and evaluated machine learning workflows to identify content at risk of declining search visibility, analyzing approximately 17 months of daily search-performance data across 70+ clients using Python, Pandas, DuckDB, and Scikit-learn.
- Engineered classification features from impressions, clicks, search visibility, query breadth, position volatility, and content-level signals; developed declining-content detection models achieving 78% accuracy in one model evaluation.
- Worked with a pseudonymized warehouse dataset containing ~81.8M rows, including a 78.8M-row daily fact table, using a star-schema structure with salted, namespaced, fingerprinted hash keys for large-scale ML and analytics workflows.

RESEARCH & TECHNICAL PUBLICATIONS

Predicting Organic-Search Visibility Decline for Content Review Prioritization [Live] | [GitHub]
Research Paper / Publication — FlyRank AI ML Engineering Internship | 2026

- Developed and evaluated a Logistic Regression model to prioritize content at risk of organic-search visibility decline using approximately 79M rows of pseudonymized search-performance data across 70+ clients.
- Achieved 78% Precision@50, compared with 92% for the heuristic baseline, and documented the 14-percentage-point performance gap through comparative evaluation and error analysis.
- Applied client-grouped validation with GroupShuffleSplit, feature analysis, leakage checks, and ranked review recommendations to assess model reliability and cross-client generalization.

PROJECTS

Aeterna AI – AI Interior Design Platform [Live] | [GitHub] *React · FastAPI · SegFormer · MiDaS · Stable Diffusion*

- Engineered a full-stack AI platform that transforms user-uploaded room images into personalized interior redesigns through an automated ML inference pipeline.
- Designed a 3-layer Preference ANN with LayerNorm and 5 output heads, encoding room type, style, density, lighting, and rating signals into a 32-dimensional preference representation.
- Integrated SegFormer, MiDaS, preference learning, and Stable Diffusion into a unified pipeline for segmentation, depth estimation, personalization, and image generation.

AI Medical Diagnostics Platform [Live] | [GitHub] *React · Flask · TensorFlow · CNN · OpenCV · Vercel*

- Built a full-stack diagnostic platform combining Pneumonia Detection and Skin Cancer Classification through a React dashboard and Flask REST API.
- Trained and deployed CNN models for chest X-ray and dermoscopic image classification, achieving approximately 90% classification accuracy.
- Added confidence-score visualization to make model predictions easier to interpret.

AI Research Critic [Live] | [GitHub] *Generative AI, LLMs, Python, Gemini API, AI Agents*

- Developed an LLM-based research-critique agent that evaluates ML/AI research ideas for hypothesis quality, data leakage, dataset limitations, methodology, baselines, evaluation metrics, and experimental design.
- Designed a structured evaluation workflow to identify methodological weaknesses, assess experimental rigor, and support research idea refinement before implementation.

SKILLS

Programming Languages: Python, SQL, JavaScript
Machine Learning: Scikit-learn, PyTorch, TensorFlow, Keras, Classification, Predictive Modeling, Feature Engineering, Model Evaluation, Convolutional Neural Networks (CNNs)
Generative AI & LLMs: Large Language Models (LLMs), Generative AI, AI Agents, Prompt Engineering, Retrieval-Augmented Generation (RAG), Stable Diffusion, Gemini API
Computer Vision: SegFormer, MiDaS, OpenCV, CNNs
Data Engineering & Analysis: Pandas, DuckDB, SQL, MySQL, MongoDB, Data Modeling, Power BI, Matplotlib, Seaborn
Model Deployment & APIs: FastAPI, Flask, Streamlit, REST APIs, React.js, Postman
MLOps & Tools: Git, GitHub, GitHub Actions (CI/CD), Jupyter Notebook, Google Colab, Anaconda, VS Code

CERTIFICATIONS

Pak Angels Gen AI Training Program — Top Performer | View Credential | Pak Angels & Aspire Pakistan | October 2026
Completed a six-week program covering Generative AI, prompt engineering, RAG, AI agents, multi-agent systems, vibe coding, and workflow automation.